Coding & App Development for Beginners

Chapter 2

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Worksheet: Chapter 2 - Getting Started with Scratch (Block-Based Coding)

This worksheet is designed for high school students in the course “Coding & App Development for Beginners.” It covers key concepts from Chapter 2 on Scratch, including the interface, coding basics, project creation, and community engagement. Complete all sections to reinforce your learning. Have fun and be creative!

Part 1: Understanding the Scratch Interface (10 points)

Instructions: Match each Scratch interface component to its correct description by writing the corresponding letter next to the number. (2 points each)

- Stage Answer: _____
- Blocks Palette Answer: _____
- Sprite List Answer: _____
- Coding Area Answer: _____
- Costumes Tab Answer: _____
- A. The area where you drag and snap blocks together to create code for your sprites. B. A section showing different looks or poses for a sprite, like a wardrobe. C. The section displaying all characters or objects in your project. D. The “TV screen” of your project where actions and animations play out. E. The toolbox containing colorful blocks categorized by function (like Motion or Looks).

Part 2: Coding Concepts in Scratch (15 points)

Instructions: Fill in the blanks with the correct term (Loop, Conditional, Variable) to complete the sentences about core coding concepts. (3 points each)

- A _________ is used to repeat actions multiple times, such as making a sprite dance back and forth continuously with a “forever” block. Answer: ______________________
- A _________ stores data like a score or counter that can change during a project, useful for tracking points in a game. Answer: ______________________
- A _________ allows your project to make decisions, like using an “if-else” block to say something only when a sprite is clicked. Answer: ______________________
- Using a _________ in Scratch can save time by avoiding the need to repeat the same blocks manually over and over. Answer: ______________________
- Combining a _________ with a sensing block can make a sprite react differently based on user input, like mouse clicks or key presses. Answer: ______________________

Part 3: Hands-On Coding Challenge (30 points)

Instructions: Create a simple Scratch project following the steps below. Write down the sequence of blocks you used for each sprite. Test your project and answer the reflection questions. (15 points for code sequence, 15 points for reflection)

Project: Interactive Pet Sprite

- Choose a sprite (like a dog or cat) and a backdrop.
- Make the sprite move to the center when the green flag is clicked.
- Use a "forever" loop to make it glide slowly left and right across the stage.
- Add a conditional: When the sprite is clicked, it should say something funny (like "Hey, that tickles!") for 2 seconds.
- Create a variable called "Pets" to count how many times the sprite is clicked, and display the count after each click.

Code Sequence for Sprite (write the blocks in order):

-
-
-
-
-

(Continue if needed) 6. ____ 7. ____

Reflection Questions:

- What was the easiest part of coding this project? (3 points)
- What was one challenge you faced, and how did you solve it (or plan to)? (6 points)
- How could you improve this project with one additional feature (like a sound or costume change)? (6 points)

Part 4: Project Analysis - Pong Game (20 points)

Instructions: Answer the following questions based on the Pong game project from Chapter 2. (5 points each)

- What is the purpose of the "if on edge, bounce" block for the ball sprite in Pong?
- How do variables like "Score1" and "Score2" enhance the gameplay experience?
- Why are conditional blocks important for detecting when the ball touches a paddle or scores a point?
- Suggest one way to customize the Pong game to make it more exciting (e.g., sprites, speed, or sounds).

Part 5: Community and Sharing (15 points)

Instructions: Answer the following short-answer questions about sharing and collaborating on Scratch. (5 points each)

- What is one benefit of sharing your Scratch project with the online community?
- What does "remixing" mean in Scratch, and why should you give credit to the original creator?
- Name one safety tip to follow when interacting with others on Scratch.

Bonus Creative Task (10 extra points)

Instructions: Brainstorm an idea for a new Scratch project (like a game, story, or animation) that combines at least two concepts from this chapter (e.g., loops and variables). Describe your idea in 3-5 sentences, including the main goal and features.

My Scratch Project Idea: